Early-Stage Pancreatic Cancer Detection

Using Random Forest Classification on Clinical Urinary Biomarker Data

A Data Mining Course Project Report

Department of Computer Science and Engineering (AI/ML)

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Keywords: Random Forest, Pancreatic Cancer, Urinary Biomarkers, SMOTE, Class Imbalance, Medical Data Mining

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) has one of the lowest five-year survival rates of any cancer because it is almost always diagnosed late, after the window for curative surgery has closed, and no low-cost screening test exists for the general population. A non-invasive, urine-based biomarker test could close this gap. This project builds an end-to-end machine learning pipeline that classifies patients as Control (healthy), Benign (non-cancerous pancreatic or hepatobiliary disease), or PDAC from urinary and plasma biomarkers (CA19-9, creatinine, LYVE1, REG1B, TFF1, REG1A) plus age, sex, and sample-collection metadata.

Random Forest is the sole data mining technique, with no other model families and no deep learning. The pipeline addresses target leakage (two fields only known after diagnosis are excluded), class imbalance (SMOTE applied inside each cross-validation fold plus class weighting), and hyperparameter selection (GridSearchCV over stratified 5-fold cross-validation, optimizing macro-F1). Quality is reported with accuracy, precision, recall, F1-score, and one-vs-rest ROC-AUC, with particular attention to recall because a missed PDAC case is the costliest error for a screening tool.

The model reaches 91.5% test accuracy and a macro ROC-AUC of 0.983 on a held-out partition never used for training or tuning. PDAC is never mistaken for a healthy Control, and residual error is concentrated on the Benign-versus-cancer boundary. Gini-based feature importance identifies REG1B, TFF1, and plasma CA19-9 as the most informative biomarkers, consistent with the clinical literature. The report covers the literature, the pipeline architecture, key code with its output, results, limitations, and future work.

1. Introduction

Pancreatic cancer remains one of the most lethal malignancies in oncology. The overwhelming majority of cases are diagnosed after the disease has progressed beyond the point of curative surgical resection, which is the single largest contributor to its poor prognosis. Unlike cancers with established screening protocols (e.g., breast, cervical, or colorectal cancer), no low-cost, widely deployable screening test exists for pancreatic cancer in the general population. Because early-stage PDAC is frequently asymptomatic, a non-invasive biomarker-based test capable of flagging at-risk individuals before symptom onset would represent a meaningful clinical advance.

Urine is an attractive medium for such a test: collection is non-invasive, inexpensive, and does not require specialized clinical infrastructure. Prior clinical research has identified a small panel of urinary proteins — including LYVE1, REG1B, and TFF1 — that, in combination with the established serum marker CA19-9 and a creatinine-based normalization control, show measurable diagnostic value for distinguishing PDAC patients from healthy controls and from patients with benign pancreatic or hepatobiliary conditions. This project builds a supervised classification pipeline around exactly this biomarker panel, using Random Forest as the sole data mining technique, and evaluates how well such a panel can be exploited by a tree-ensemble classifier for three-way diagnostic classification.

Section 2 surveys the medical and machine learning literature behind the design. Section 3 describes the methodology and pipeline. Section 4 shows key code and its console output. Section 5 presents results and compares them with prior literature. Section 6 covers limitations and Section 7 concludes. The first pages of the two principal papers follow the references.

2. Literature Survey

This section reviews the clinical research on urinary biomarker panels for pancreatic cancer and the machine learning literature on Random Forests, the project's sole classification technique, plus a few supporting references.

2.1 Medical Literature: Urinary Biomarker Panels for Pancreatic Cancer

Debernardi et al. (2020) [1] report a case–control study of 590 retrospectively collected urine specimens from six centres: 183 controls, 208 patients with benign hepatobiliary disease (119 with chronic pancreatitis) and 199 patients with PDAC (102 stage I–II, 97 stage III–IV). The authors assayed three urinary proteins — LYVE1, TFF1 and REG1B, which replaced REG1A from the group's earlier panel — and combined them with urinary creatinine and age in PancRISK, a logistic regression model that labels each patient's risk of PDAC as “normal” or “elevated”. All three proteins were significantly higher in PDAC urine than in benign and control samples at every stage (Kruskal–Wallis, $p < 0.0001$); substituting REG1B for REG1A improved detection of resectable (stage I–II) disease; and adding plasma CA19-9 improved the panel further (AUC 0.992 for stage I–II PDAC versus controls). The authors name the small number of stage I–IIA samples ($n = 27$) and the variability of commercial ELISA kits as their main limitations.

This study shaped the present project in three ways. First, it supplied the core feature set: the urinary proteins LYVE1, REG1B and TFF1, urinary creatinine, age and plasma CA19-9. This project also keeps REG1A, the marker that REG1B replaced in [1], and adds sex, cohort and sample-origin metadata. Second, its three diagnostic groups — control, benign hepatobiliary disease and PDAC — define the three-class problem addressed here; [1] compares PDAC with controls and with benign samples separately, so a single three-way classifier extends its design rather than replicating it. Third, PancRISK is itself a multivariable logistic regression, which motivated testing whether a non-linear tree ensemble can extract more from the same variables.

2.2 Machine Learning Literature: Random Forest Classification

Breiman (2001) [2] introduced random forests: combinations of tree predictors in which each tree depends on the values of a random vector sampled independently, and with the same distribution, for all trees in the forest. The paper shows that the generalization error converges almost surely to a limit as the number of trees grows, and that it depends on the strength of the individual trees and on the correlation between them. Selecting a random subset of the features at each node gives error rates that compare favorably with AdaBoost while being more robust to noise and faster to train, and “internal estimates” based on out-of-bag data are used to monitor error, strength and correlation and to measure variable importance.

This work is the direct algorithmic foundation of the present project in two ways. First, the assignment specifies Random Forest as the sole permitted data mining technique (no ensembling with other model families, no deep learning); Breiman's formulation is the algorithm implemented by scikit-learn's RandomForestClassifier. His result that error depends on tree strength and tree correlation is why the hyperparameter grid in Section 3.3 varies tree depth, leaf size and the number of features per split. Second, his idea that variable importance comes from the forest itself is what lets this project answer its clinical question (“which biomarkers matter most?”) with the classifier alone. One caveat: Breiman defines importance by randomly permuting a variable's values in the out-of-bag data and measuring the loss in accuracy, whereas scikit-learn's feature_importances_, used in Section 5.3, is the related but distinct Gini (mean decrease in impurity) measure; permutation importance is therefore listed as future work. Section 3.3 illustrates the ensemble mechanism.

2.3 Supporting Literature

Two further references informed secondary design decisions. Chawla et al. (2002) [3] introduced the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic examples of the minority class along line segments joining existing minority-class nearest neighbors, rather than simply duplicating them. SMOTE is used in this project's training pipeline, applied only within each cross-validation fold, to counteract the mild class imbalance across the Control, Benign, and PDAC groups. Siegel et al. (2023) [4] report updated cancer incidence and survival statistics, including the consistently poor five-year survival rate for pancreatic cancer relative to other major cancers; these figures motivate the clinical framing given in Section 1 and Section 6.

3. Proposed Methodology and Pipeline Architecture

The complete pipeline consists of five stages, illustrated in Figure 1: data loading, preprocessing, model training (with class-imbalance handling and hyperparameter tuning folded in), evaluation, and inference on new samples. Each stage is implemented as an independent, re-runnable script, and Random Forest is used as the sole data mining technique across the entire pipeline — no ensembling with other model families and no deep learning are used at any stage.

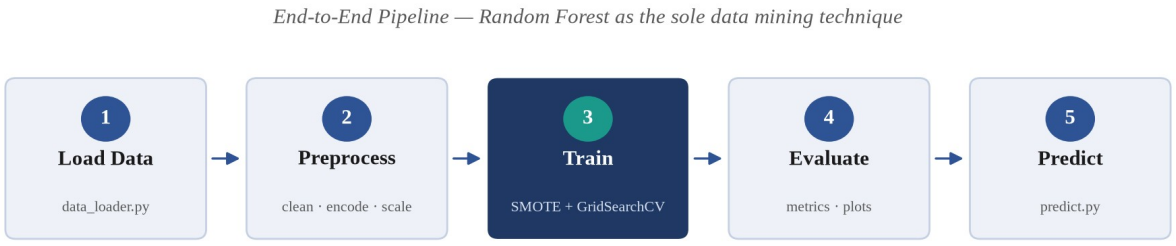


Figure 1. *End-to-end pipeline architecture. Random Forest (stage 3) is the only data mining technique used.*

3.1 Preprocessing

- Target-leakage removal: two raw fields — disease stage and benign-condition sub-type — are dropped before modeling, since both are only recorded after a diagnosis has already been made and would therefore leak the target label into the model rather than reflect genuine predictive signal.
- Missing-value imputation: CA19-9 and REG1A, which have real assay-driven missingness, are median-imputed using statistics computed on the training fold only.
- Categorical encoding: sex, patient cohort, and sample origin are one-hot encoded.
- Feature scaling: numeric features are standardized (zero mean, unit variance).
- Stratified 80/20 train–test split, preserving class proportions in both partitions.

3.2 Class Imbalance Handling

Following Chawla et al. [3], SMOTE is applied to oversample the minority classes during training. Critically, SMOTE is embedded inside an imblearn pipeline so that resampling is refit independently on each cross-validation fold's training portion; this prevents synthetic samples generated from one fold from leaking into another fold's validation data, which would otherwise produce an optimistic bias in cross-validated performance estimates. Cost-sensitive class weighting (class_weight = 'balanced') is additionally applied within the Random Forest itself as a complementary, low-cost second layer of imbalance handling.

3.3 Model: Random Forest Classification

Following Breiman [2], a Random Forest classifier is trained as the sole data mining technique specified for this project. A Random Forest is an ensemble of decision trees: each tree is trained on an independent bootstrap sample drawn (with replacement) from the training data, and at every split each tree considers only a random subset of the available features rather than all of them. This double randomization — over both samples and features — decorrelates the individual trees, so that their errors are largely independent of one another. At prediction time, every tree casts a vote for a diagnostic class, and the forest's final prediction is the majority vote across all trees, as illustrated in Figure 2.

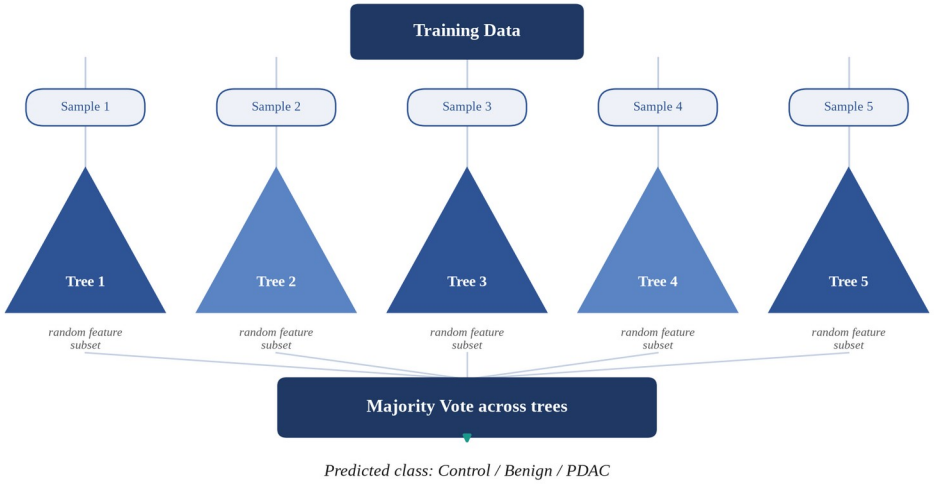


Figure 2. *Random Forest ensemble mechanism: bootstrap-resampled training subsets, independently grown trees with randomized feature subsets, combined by majority vote.*

Hyperparameters are tuned via GridSearchCV over stratified 5-fold cross-validation, optimizing macro-averaged F1-score — which weights all three diagnostic classes equally, appropriate given that missing a minority PDAC case is clinically more costly than an error on the majority classes. The search space covers the number of trees (n_estimators ∈ {100, 200, 400}), maximum tree depth (max_depth ∈ {None, 5, 10}), minimum leaf size (min_samples_leaf ∈ {1, 2, 4}), and the number of features considered per split (max_features ∈ {sqrt, log2}) — 54 candidate combinations evaluated across 5 folds, for 270 total model fits. The best configuration (n_estimators = 200, max_depth = 10, min_samples_leaf = 1, max_features = 'sqrt') achieved a cross-validated macro-F1 of 0.939. The complete training procedure, as actually executed, is listed in Section 4.

3.4 Evaluation Metrics

Model quality is quantified with five complementary metrics, each capturing a different aspect of classification performance:

- Accuracy = (TP + TN) / Total — the overall fraction of correct predictions; can be misleading on its own under class imbalance.
- Precision = TP / (TP + FP) — of samples predicted as a given class, the fraction that truly belong to it.
- Recall (sensitivity) = TP / (TP + FN) — of samples that truly belong to a class, the fraction correctly identified; the critical metric for a screening tool, since a false negative (a missed PDAC case) is the costliest possible error.
- F1-score = 2 · Precision · Recall / (Precision + Recall) — the harmonic mean of precision and recall; reported here as macro-F1, which averages the F1-score across all three classes with equal weight.
- ROC-AUC (one-vs-rest) — sweeps every possible decision threshold and plots the True Positive Rate against the False Positive Rate for each class treated as positive-vs-rest; the Area Under the Curve (0–1) summarizes ranking quality independent of any single threshold, with 0.5 corresponding to random guessing and 1.0 to perfect separation.

The held-out test partition is used exclusively for final evaluation and is never accessed during training or hyperparameter selection, so the metrics reported in Section 5 reflect genuine generalization performance rather than training fit. Feature importance is read directly from the trained forest as scikit-learn's Gini-based mean decrease in impurity, a by-product of training that is closely related to, but distinct from, the permutation-based importance defined in [2].

4. Implementation: Code and Execution Output

This section reproduces the key parts of the code that trains and evaluates the model, together with the real console output of the run behind the results in Section 5. The full scripts are in src/ of the project repository; each pipeline stage is an independent, re-runnable script.

4.1 Model Training Script (train.py)

train.py loads the preprocessed features, builds an imblearn pipeline chaining SMOTE and the Random Forest classifier (so resampling is refit per cross-validation fold rather than once globally), runs the GridSearchCV hyperparameter search described in Section 3.3, and persists the best model together with the untouched test split for later evaluation.

```
PARAM_GRID = {
    "rf__n_estimators": [100, 200, 400],
    "rf__max_depth": [None, 5, 10],
    "rf__min_samples_leaf": [1, 2, 4],
    "rf__max_features": ["sqrt", "log2"],
}
def build_pipeline() -> ImbPipeline:
    return ImbPipeline(
        steps=[
            ("smote", SMOTE(random_state=RANDOM_STATE)),
            (
                "rf",
                RandomForestClassifier(
                    random_state=RANDOM_STATE,
                    class_weight="balanced",
                    n_jobs=-1,
                ),
            ),
        ]
    )
def train_model(X_train, y_train):
    cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)
    pipeline = build_pipeline()

    search = GridSearchCV(
        estimator=pipeline,
        param_grid=PARAM_GRID,
        scoring="f1_macro",
        cv=cv,
        n_jobs=-1,
        verbose=1,
    )
    search.fit(X_train, y_train)
    return search
```

Code Listing 1. train.py — key excerpt (hyperparameter grid, SMOTE + Random Forest pipeline, and GridSearchCV). Imports, docstrings, and model-persistence code are omitted; the complete file is src/train.py.

4.2 Evaluation Script (evaluate.py)

evaluate.py loads the model and test split saved by train.py, computes accuracy, the per-class classification report, and macro ROC-AUC on the held-out test set only, and renders the confusion matrix, ROC curve, and feature-importance figures reproduced in Section 5.

```
def compute_metrics(model, X_test, y_test) -> dict:
    y_pred = model.predict(X_test)
    y_proba = model.predict_proba(X_test)

    report = classification_report(y_test, y_pred, target_names=CLASS_NAMES, output_dict=True)
    accuracy = accuracy_score(y_test, y_pred)

    # Multi-class ROC-AUC needs one-hot ("binarized") true labels.
    y_test_bin = label_binarize(y_test, classes=[0, 1, 2])
    roc_auc_macro = roc_auc_score(y_test_bin, y_proba, average="macro", multi_class="ovr")

    metrics = {
        "accuracy": accuracy,
        "roc_auc_macro_ovr": roc_auc_macro,
        "classification_report": report,
    }
    return metrics, y_pred, y_proba, y_test_bin
```

Code Listing 2. evaluate.py — key excerpt (metric computation on the held-out test set). Plotting and I/O code are omitted; the complete file is src/evaluate.py.

4.3 Captured Execution Output

Condensed console output of python train.py and python evaluate.py (data-loading and file-save messages omitted). All randomness uses a fixed random_state, so re-running reproduces this output.

```
[train] Loading and preprocessing data...
[train] Running GridSearchCV (SMOTE + RandomForest, 5-fold CV, scoring=f1_macro)...
Fitting 5 folds for each of 54 candidates, totalling 270 fits
[train] Best CV macro-F1: 0.9393
[train] Best hyperparameters: {'rf__max_depth': 10, 'rf__max_features': 'sqrt',
'rf__min_samples_leaf': 1, 'rf__n_estimators': 200}
[evaluate] Accuracy: 0.9153
[evaluate] Macro ROC-AUC (OvR): 0.9834
[evaluate] Classification report:
              precision    recall  f1-score   support

     Control       1.00      0.89      0.94        37
      Benign       0.83      0.95      0.89        41
        PDAC       0.95      0.90      0.92        40

    accuracy                           0.92       118
[evaluate] Top 5 most important biomarkers:
  REG1B: 0.2614
  TFF1: 0.2437
  plasma_CA19_9: 0.1852
```

Output 2. Captured console output of `python evaluate.py`.

5. Results and Discussion

The tuned Random Forest achieved a best cross-validated macro-F1 of 0.939 during grid search. On the held-out test partition (118 samples, never used in training or tuning), the model achieved the following per-class performance:

Table 1. Per-class precision, recall, F1-score, and support on the held-out test set.

Class	Precision	Recall	F1-score	Support
Control	1.000	0.892	0.943	37
Benign	0.830	0.951	0.886	41
PDAC	0.947	0.900	0.923	40

Overall test accuracy was 91.5%, with a macro-averaged one-vs-rest ROC-AUC of 0.983 and a macro-F1 of 0.917.

5.1 Confusion Matrix Analysis

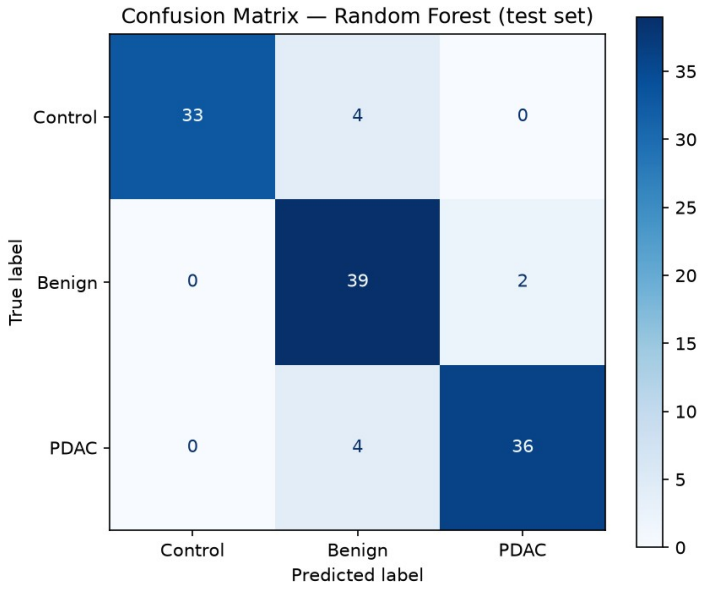


Figure 3. *Confusion matrix on the held-out test set (rows = true label, columns = predicted label).*

The confusion matrix's diagonal represents correct predictions; off-diagonal cells represent misclassifications. Two properties stand out. First, PDAC is never confused with Control (0 of 40 PDAC samples predicted as Control) — the clinically critical distinction for an early-screening use case, since it means the model never mistakes a cancer patient for a fully healthy one. Second, the errors that do occur are asymmetric in clinical cost: 4 PDAC samples were predicted as Benign

(false negatives), which is the costliest error type in a screening context because a missed PDAC case delays treatment; by contrast, 2 Benign samples were predicted as PDAC (false positives), a comparatively low-cost error since it triggers additional, safe clinical workup rather than a missed diagnosis. Most residual error sits between the Benign class and the other two classes, plausibly because benign hepatobiliary disease, including chronic pancreatitis, is a recognized source of diagnostic confusion with PDAC owing to overlapping symptoms [1].

5.2 ROC Curve Analysis

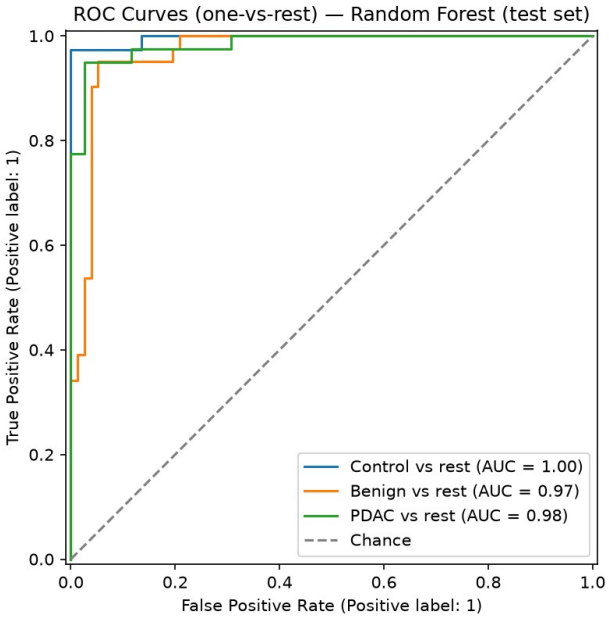


Figure 4. *One-vs-rest ROC curves for each diagnostic class on the held-out test set.*

Each curve treats one diagnostic class as positive against the other two combined, sweeping the decision threshold from 0 to 1. All three curves hug the top-left corner, indicating strong class separation at almost every threshold, not merely the conventional 0.5 cutoff. The Control class reaches a per-class AUC of 1.00, meaning the model perfectly ranks healthy patients above non-healthy ones by predicted probability across the test set. Benign has the lowest per-class AUC at 0.97, mirroring the confusion-matrix finding that Benign is the hardest class to isolate; PDAC sits at 0.98. The resulting macro-averaged AUC of 0.983 means that, averaged across the three classes, a randomly chosen true-positive sample is ranked above a randomly chosen true-negative sample 98.3% of the time — a strong result given the modest training-set size.

5.3 Feature Importance Analysis

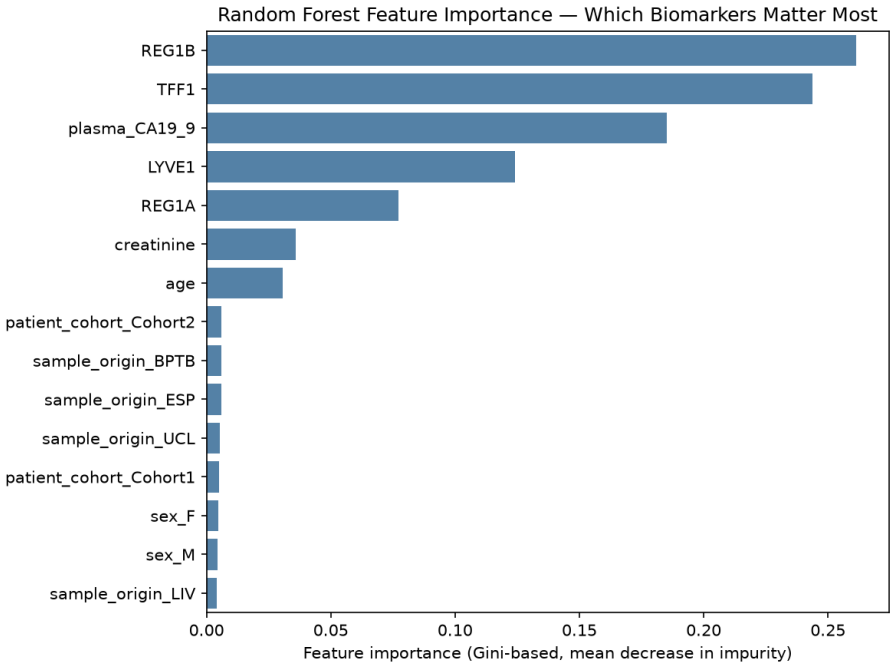


Figure 5. *Random Forest Gini-based feature importance, ranking which biomarkers the trained model relies on most.*

Feature-importance analysis (scikit-learn's Gini-based mean decrease in impurity) ranked REG1B (0.261), TFF1 (0.244), and plasma CA19-9 (0.185) as the three most predictive features, followed by LYVE1 (0.124) and REG1A (0.077). Age and the sex, cohort and sample-origin metadata together contributed only about 7% of total importance (age about 3%, every other variable below 1%), which is reassuring, since it suggests the model relies on the biomarker measurements rather than on collection-site artifacts or demographic shortcuts. Gini importance can overstate continuous variables, so the ranking should be confirmed with permutation importance, the measure defined in [2].

5.4 Comparison with Prior Literature

Debernardi et al. [1] do not rank their markers by importance, so this comparison is qualitative. They report that LYVE1, REG1B and TFF1 are all significantly higher in PDAC urine than in benign and control samples, that substituting REG1B for REG1A improved detection of stage I–II disease, and that plasma CA19-9 adds complementary information to the urinary panel. Table 2 sets the Random Forest's Gini importances beside these findings:

Table 2. This project's Random Forest feature importances alongside the findings Debernardi et al. [1] report for the same biomarkers.

Biomarker	Gini importance (this project)	Finding in Debernardi et al. [1]
REG1B	0.261 (rank 1)	Higher in PDAC than in benign and control urine ($p < 0.0001$); replaced REG1A in the improved panel
TFF1	0.244 (rank 2)	Higher in PDAC than in benign and control urine ($p < 0.0001$)
Plasma CA19-9	0.185 (rank 3)	Complementary to the urinary panel; combined AUC 0.992 for stage I–II PDAC versus controls
LYVE1	0.124 (rank 4)	Higher in PDAC than in benign and control urine ($p < 0.0001$)
REG1A	0.077 (rank 5)	Marker of the group's earlier panel; superseded by REG1B

The two sets of results point the same way: the urinary proteins and CA19-9 that [1] finds elevated or complementary in PDAC are the variables the Random Forest relies on most, while age and the cohort and site metadata carry little weight. This is consistent with, but does not independently confirm, the clinical findings: importance measures how much the model uses a variable, not whether that variable is clinically valid.

6. Limitations

- Sample size (n = 590) is modest by machine learning standards, though consistent with typical clinical biomarker study cohorts such as [1].
- Evaluation relies on a single stratified train/test split; repeated cross-validated reporting would tighten confidence intervals on the metrics in Section 5.
- Gini-based feature importance can be biased toward high-cardinality or continuous features relative to categorical ones; permutation importance would provide a useful cross-check.
- No external, site-held-out validation cohort was used; given that samples originate from multiple collection sites, leave-one-cohort-out validation would better estimate cross-site generalization.
- The clinically most important group, early-stage PDAC, is small: [1] itself had only 27 stage I–IIA samples, so any early-detection claim rests on few cases.
- The biomarkers are measured with commercial ELISAs whose performance varies between kit versions [1]; assay variability between sites could affect any model trained on these values.

7. Conclusion and Future Work

This project demonstrates that a Random Forest classifier, trained on the urinary and plasma biomarker panel established by Debernardi et al. [1] and built according to the ensemble methodology of Breiman [2], can distinguish Control, Benign, and PDAC diagnostic groups with 91.5% accuracy and a macro-averaged ROC-AUC of 0.983. Confusion-matrix and ROC analysis show that the model separates PDAC from Control almost perfectly, with the majority of residual error concentrated on the harder Benign-vs-cancer boundary. Feature-importance analysis identifies REG1B, TFF1, and CA19-9 as the most clinically informative biomarkers, the proteins and plasma marker that [1] reports as elevated in, or complementary for, PDAC. Future work should prioritize (i) permutation and SHAP-based importance as a cross-check on the Gini-based ranking, (ii) leave-one-cohort-out validation across collection sites, (iii) a focused evaluation on early-stage-only PDAC (Stage I/II) detection, which is the highest-value and most clinically demanding use case, and (iv) probability calibration if model outputs are to inform an actual clinical decision threshold.

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Research Paper A: A combination of urinary biomarker panel and PancRISK score for earlier detection of pancreatic cancer: A case–control study,

S. Debernardi, H. O'Brien, A. S. Algahmdi, et al., "A combination of urinary biomarker panel and PancRISK score for earlier detection of pancreatic cancer: A case–control study," PLOS Medicine, vol. 17, no. 12, e1003489, 2020. First page shown; click the title above to open the complete paper (https://doi.org/10.1371/journal.pmed.1003489).

PLOS MEDICINE

RESEARCH ARTICLE

A combination of urinary biomarker panel and PancRISK score for earlier detection of pancreatic cancer: A case–control study

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Citation: Debernardi S, O'Brien H, Algahmdi AS, Malats N, Stewart GD, Plješa-Ercegovac M, et al. (2020) A combination of urinary biomarker panel and PancRISK score for earlier detection of pancreatic cancer: A case–control study. PLoS Med 17(12): e1003489. https://doi.org/10.1371/journal.pmed.1003489

Academic Editor: Michele T. Yip-Schneider, Indiana University School of Medicine, UNITED STATES

Received: April 30, 2020

Accepted: November 19, 2020

Published: December 10, 2020

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Data Availability Statement: All relevant data are within the manuscript and its Supporting Information files.

Funding: The study was funded by Pancreatic Cancer Research Fund (PCRF) (www.pcrf.org.uk). The award was assigned to TCJ. The DIAMOND study is part supported by infrastructure from the Cambridge NIHR Biomedical Campus and CRUK Cambridge Cancer Centre. The salary of SD was

Abstract

Background

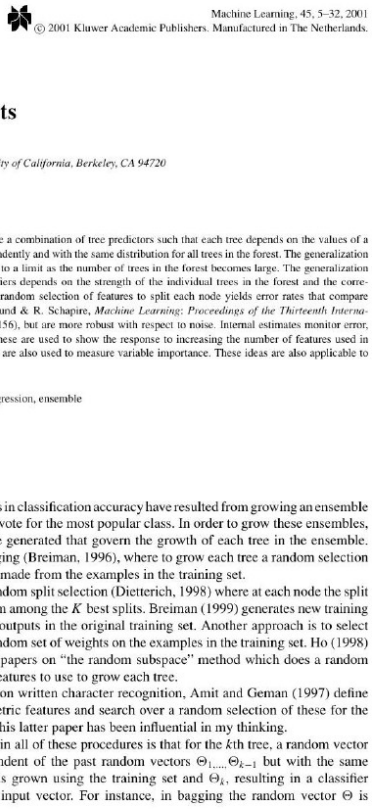
Pancreatic ductal adenocarcinoma (PDAC) is one of the deadliest cancers, with around 9% of patients surviving >5 years. Asymptomatic in its initial stages, PDAC is mostly diagnosed late, when already a locally advanced or metastatic disease, as there are no useful biomarkers for detection in its early stages, when surgery can be curative. We have previously described a promising biomarker panel (LYVE1, REG1A, and TFF1) for earlier detection of PDAC in urine. Here, we aimed to establish the accuracy of an improved panel, including REG1B instead of REG1A, and an algorithm for data interpretation, the PancRISK score, in additional retrospectively collected urine specimens. We also assessed the complementarity of this panel with CA19-9 and explored the daily variation and stability of the biomarkers and their performance in common urinary tract cancers.

Methods and findings

Clinical specimens were obtained from multiple centres: Barts Pancreas Tissue Bank, University College London, University of Liverpool, Spanish National Cancer Research Center, Cambridge University Hospital, and University of Belgrade. The biomarker panel was assayed on 590 urine specimens: 183 control samples, 208 benign hepatobiliary disease samples (of which 119 were chronic pancreatitis), and 199 PDAC samples (102 stage I–II and 97 stage III–IV); 50.7% were from female individuals. PDAC samples were collected

Research Paper B: Random Forests,

L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001. First page shown; click the title above to open the complete paper (https://doi.org/10.1023/A:1010933404324).



Machine Learning, 45, 5–32, 2001
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Random Forests

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Editor: Robert E. Schapire

Abstract. Random forests are a combination of tree predictors such that each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest. The generalization error for forests converges a.s. to a limit as the number of trees in the forest becomes large. The generalization error of a forest of tree classifiers depends on the strength of the individual trees in the forest and the correlation between them. Using a random selection of features to split each node yields error rates that compare favorably to Adaboost (Y. Freund & R. Schapire, *Machine Learning: Proceedings of the Thirteenth International conference*, ∗ ∗ ∗, 148–156), but are more robust with respect to noise. Internal estimates monitor error, strength, and correlation and these are used to show the response to increasing the number of features used in the splitting. Internal estimates are also used to measure variable importance. These ideas are also applicable to regression.

Keywords: classification, regression, ensemble

1. Random forests

1.1. Introduction

Significant improvements in classification accuracy have resulted from growing an ensemble of trees and letting them vote for the most popular class. In order to grow these ensembles, often random vectors are generated that govern the growth of each tree in the ensemble. An early example is bagging (Breiman, 1996), where to grow each tree a random selection (without replacement) is made from the examples in the training set.

Another example is random split selection (Dietterich, 1998) where at each node the split is selected at random from among the K best splits. Breiman (1999) generates new training sets by randomizing the outputs in the original training set. Another approach is to select the training set from a random set of weights on the examples in the training set. Ho (1998) has written a number of papers on "the random subspace" method which does a random selection of a subset of features to use to grow each tree.

In an important paper on written character recognition, Amit and Geman (1997) define a large number of geometric features and search over a random selection of these for the best split at each node. This latter paper has been influential in my thinking.

The common element in all of these procedures is that for the kth tree, a random vector Θ_k is generated, independent of the past random vectors $\Theta_1, \ldots, \Theta_{k-1}$ but with the same distribution; and a tree is grown using the training set and Θ_k, resulting in a classifier $h(\mathbf{x}, \Theta_k)$ where $\mathbf{x}$ is an input vector. For instance, in bagging the random vector Θ is